

Name: \_\_\_\_\_

A Level Mathematics : Pure Mathematics

Topic 2 :Algebra and functions -Partial fractions

**Date:**

**Time:**

**Total marks available:**

**Total marks achieved:** \_\_\_\_\_

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## Questions

Q1.

- (a) Express  $\frac{3}{(2x-1)(x+1)}$  in partial fractions.

(3)

When chemical *A* and chemical *B* are mixed, oxygen is produced.

A scientist mixed these two chemicals and measured the total volume of oxygen produced over a period of time.

The total volume of oxygen produced,  $V \text{ m}^3$ ,  $t$  hours after the chemicals were mixed, is modelled by the differential equation

$$\frac{dV}{dt} = \frac{3V}{(2t-1)(t+1)} \quad V \geq 0 \quad t \geq k$$

where  $k$  is a constant.

Given that exactly 2 hours after the chemicals were mixed, a total volume of  $3 \text{ m}^3$  of oxygen had been produced,

- (b) solve the differential equation to show that

$$V = \frac{3(2t-1)}{(t+1)}$$

(5)

The scientist noticed that

- there was a **time delay** between the chemicals being mixed and oxygen being produced
- there was a **limit** to the total volume of oxygen produced

Deduce from the model

- (c) (i) the **time delay** giving your answer in minutes,  
 (ii) the **limit** giving your answer in  $\text{m}^3$

(2)

**(Total for question = 10 marks)**

Q2.

$$\frac{1 + 11x - 6x^2}{(x - 3)(1 - 2x)} \equiv A + \frac{B}{(x - 3)} + \frac{C}{(1 - 2x)}$$

(a) Find the values of the constants  $A$ ,  $B$  and  $C$ .

(4)

$$f(x) = \frac{1 + 11x - 6x^2}{(x - 3)(1 - 2x)} \quad x > 3$$

(b) Prove that  $f(x)$  is a decreasing function.

(3)

**(Total for question = 7 marks)**

Q3.

The function  $f$  is defined by

$$f(x) = \frac{8x + 5}{2x + 3} \quad x > -\frac{3}{2}$$

(a) Find  $f^{-1}\left(\frac{3}{2}\right)$

(2)

(b) Show that

$$f(x) = A + \frac{B}{2x + 3}$$

where  $A$  and  $B$  are constants to be found.

(2)

The function  $g$  is defined by

$$g(x) = 16 - x^2 \quad 0 \leq x \leq 4$$

(c) State the range of  $g^{-1}$

(1)

(d) Find the range of  $f \circ g^{-1}$

(3)

**(Total for question = 8 marks)**

Q4.

The curve  $C$  with equation

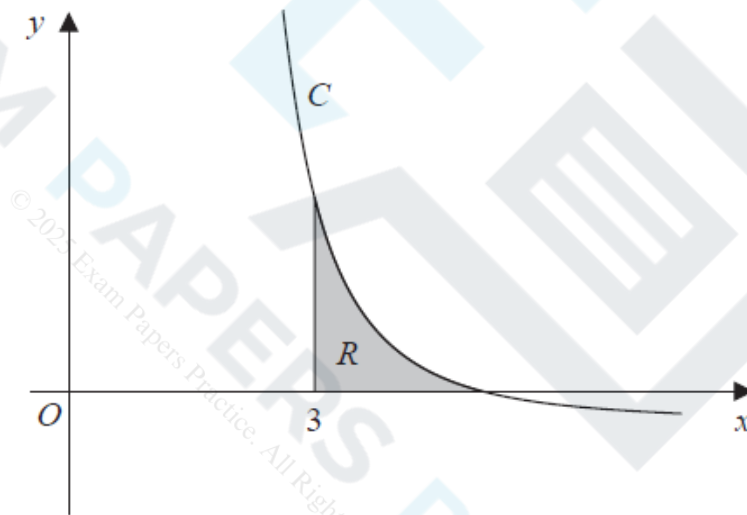
$$y = \frac{p - 3x}{(2x - q)(x + 3)} \quad x \in \mathbb{R}, x \neq -3, x \neq 2$$

where  $p$  and  $q$  are constants, passes through the point  $\left(3, \frac{1}{2}\right)$  and has two vertical asymptotes with equations  $x = 2$  and  $x = -3$

(a) (i) Explain why you can deduce that  $q = 4$

(ii) Show that  $p = 15$

(3)



**Figure 4**

Figure 4 shows a sketch of part of the curve  $C$ . The region  $R$ , shown shaded in Figure 4, is bounded by the curve  $C$ , the  $x$ -axis and the line with equation  $x = 3$

(b) Show that the exact value of the area of  $R$  is  $a \ln 2 + b \ln 3$ , where  $a$  and  $b$  are rational constants to be found.

(8)

**(Total for question = 11 marks)**

Q5.

The table below shows corresponding values of  $x$  and  $y$  for  $y = \sqrt{\frac{x}{1+x}}$

The values of  $y$  are given to 4 significant figures.

|     |        |        |        |        |        |
|-----|--------|--------|--------|--------|--------|
| $x$ | 0.5    | 1      | 1.5    | 2      | 2.5    |
| $y$ | 0.5774 | 0.7071 | 0.7746 | 0.8165 | 0.8452 |

(a) Use the trapezium rule, with all the values of  $y$  in the table, to find an estimate for

$$\int_{0.5}^{2.5} \sqrt{\frac{x}{1+x}} \, dx$$

giving your answer to 3 significant figures.

(3)

(b) Using your answer to part (a), deduce an estimate for  $\int_{0.5}^{2.5} \sqrt{\frac{9x}{1+x}} \, dx$

(1)

Given that

$$\int_{0.5}^{2.5} \sqrt{\frac{9x}{1+x}} \, dx = 4.535 \text{ to 4 significant figures}$$

(c) comment on the accuracy of your answer to part (b).

(1)

**(Total for question = 5 marks)**

Q6.

$$f(x) = \frac{50x^2 + 38x + 9}{(5x + 2)^2(1 - 2x)} \quad x \neq -\frac{2}{5} \quad x \neq \frac{1}{2}$$

Given that  $f(x)$  can be expressed in the form

$$\frac{A}{5x + 2} + \frac{B}{(5x + 2)^2} + \frac{C}{1 - 2x}$$

where  $A$ ,  $B$  and  $C$  are constants

(a) (i) find the value of  $B$  and the value of  $C$

(ii) show that  $A = 0$

(4)

(b) (i) Use binomial expansions to show that, in ascending powers of  $x$

$$f(x) = p + qx + rx^2 + \dots$$

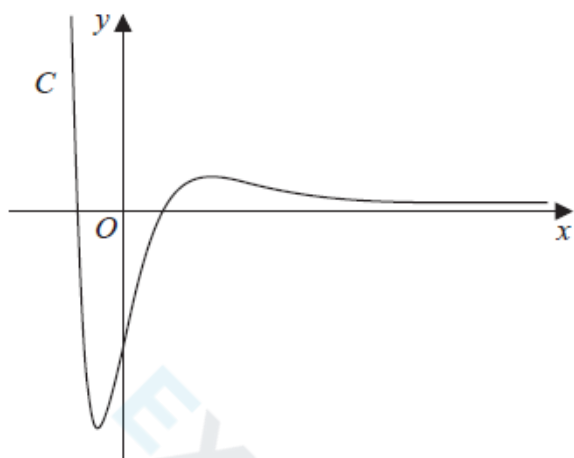
where  $p$ ,  $q$  and  $r$  are simplified fractions to be found.

(ii) Find the range of values of  $x$  for which this expansion is valid.

(7)

**(Total for question = 11 marks)**

Q7.



**Figure 2**

Figure 2 shows a sketch of the curve  $C$  with equation  $y = f(x)$  where

$$f(x) = 4(x^2 - 2)e^{-2x} \quad x \in \mathbb{R}$$

(a) Show that  $f'(x) = 8(2 + x - x^2)e^{-2x}$

(3)

(b) Hence find, in simplest form, the exact coordinates of the stationary points of  $C$ .

(3)

The function  $g$  and the function  $h$  are defined by

$$g(x) = 2f(x) \quad x \in \mathbb{R}$$

$$h(x) = 2f(x) - 3 \quad x \geq 0$$

(c) Find (i) the range of  $g$

(ii) the range of  $h$

(3)

**(Total for question = 9 marks)**